

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Data Structures Practical

Practical – 1

Roll No.: S056	Name: Shweta Singh
Class:SYIT	Batch: B1
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PART – A

Array Operations:

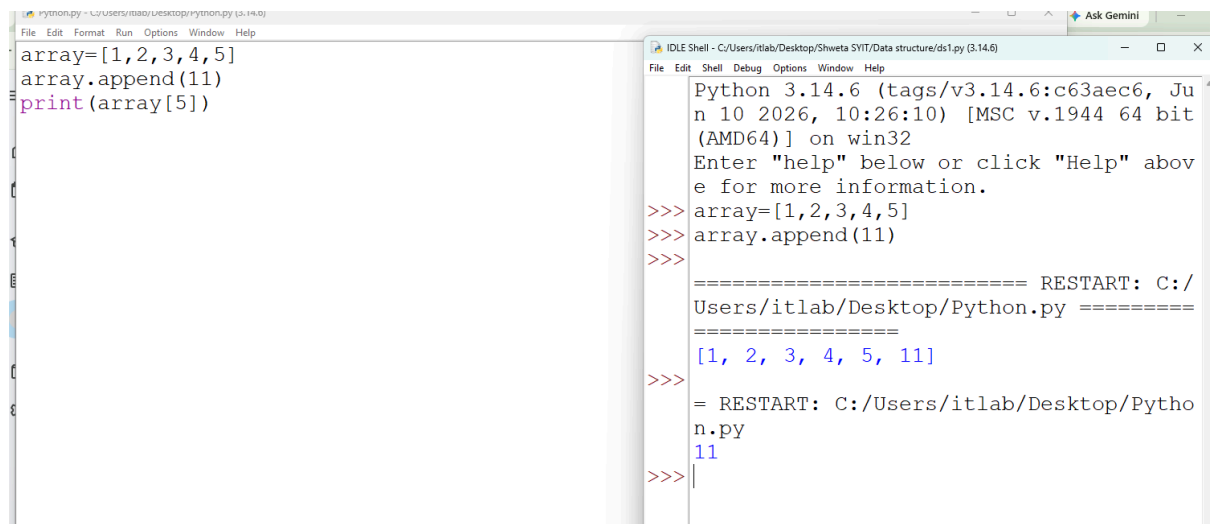
Write a program to implement basic array operations:

- a. Insert an element at a specific position in an array.
- b. Delete an element from a specific position in an array.
- c. Search for an element in an array (linear search).

PART B :

Solution:

a. Insert an element at a specific position in an array.



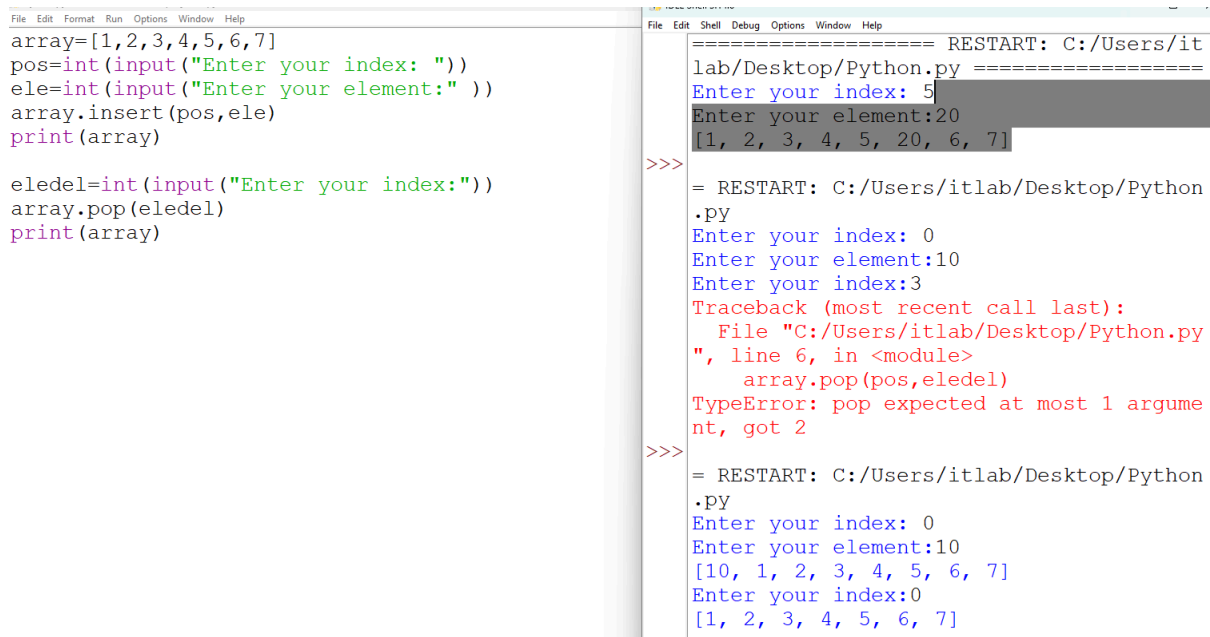
The image shows two side-by-side windows from a Python IDE. The left window, titled 'python.py', contains the following code:

```
array=[1,2,3,4,5]
array.append(11)
print(array[5])
```

The right window, titled 'IDLE Shell - C:/Users/itlab/Desktop/Shweta SYIT/Data structure/ds1.py (3.14.6)', shows the execution of the code. It displays the Python version and system information, followed by the code execution and the resulting array output:

```
>>> array=[1,2,3,4,5]
>>> array.append(11)
>>>
===== RESTART: C:/Users/itlab/Desktop/Python.py =====
[1, 2, 3, 4, 5, 11]
>>>
= RESTART: C:/Users/itlab/Desktop/Python.py
11
>>>
```

b. Delete an element from a specific position in an array.



The image shows two side-by-side windows from a Python IDE. The left window, titled 'python.py', contains the following code:

```
array=[1,2,3,4,5,6,7]
pos=int(input("Enter your index: "))
ele=int(input("Enter your element: "))
array.insert(pos,ele)
print(array)

eledel=int(input("Enter your index:"))
array.pop(eledel)
print(array)
```

The right window, titled 'IDLE Shell - C:/Users/itlab/Desktop/Shweta SYIT/Data structure/ds1.py (3.14.6)', shows the execution of the code. It displays the Python version and system information, followed by the code execution and the resulting array output:

```
>>>
===== RESTART: C:/Users/itlab/Desktop/Python.py =====
Enter your index: 5
Enter your element: 20
[1, 2, 3, 4, 5, 20, 6, 7]
>>>
= RESTART: C:/Users/itlab/Desktop/Python.py
Enter your index: 0
Enter your element: 10
Enter your index: 3
Traceback (most recent call last):
  File "C:/Users/itlab/Desktop/Python.py", line 6, in <module>
    array.pop(pos,eledel)
TypeError: pop expected at most 1 argument, got 2
>>>
= RESTART: C:/Users/itlab/Desktop/Python.py
Enter your index: 0
Enter your element: 10
[10, 1, 2, 3, 4, 5, 6, 7]
Enter your index: 0
[1, 2, 3, 4, 5, 6, 7]
```

c. Search for an element in an array (linear search).

File Edit Format Run Options Window Help

```
array=[1,2,3,4,5,6,7]
pos=int(input("Enter your index: "))
ele=int(input("Enter your element:"))
array.insert(pos,ele)
print(array)
```

File Edit Shell Debug Options Window Help

```
Python 3.14.6 (tags/v3.14.6:c63
aec6, Jun 10 2026, 10:26:10) [M
SC v.1944 64 bit (AMD64)] on wi
n32
Enter "help" below or click "He
lp" above for more information.
```

>>>

```
===== RESTART: C:
/Users/itlab/Desktop/Python.py
=====
```

```
Enter your index: 5
Enter your element:20
[1, 2, 3, 4, 5, 20, 6, 7]
```

>>>